

Clinical Pharmacokinetics: Rational Dosing and the Time Course of Drug Action

Practice Quiz Questions, Answers, and Rationales (Explanations)

Pharmacology bridges physiology, pathophysiology, and clinical practice.

Some items may have more than one answer.

CHALLENGE: Cover the options and try to answer the questions without looking at the choices. Then, explain why the other options are incorrect.

Goals of Practice Quizzes:

1. To help students identify what they know and understand by applying knowledge to clinical case vignettes.
2. To help students get used to thinking through the case vignettes that are used “Board-style” questions. NYITYCOM, COMLEX and USMLE examine students’ medical knowledge using these types of questions.

My recommendation is to highlight in the vignette the relevant information (“clues”) you need to answer the question. *Don’t let yourself get distracted by the background information unless it is relevant for answering the question.*

Where can I find the answers to the questions?

If you wish to use them, all answers can be found by **doing a “search” of the Notes Handout** or the Examination and Board Review resources designated on the CPG (links are provided). The handout and textbook readings are not a required, but you may find one or the other to be VERY helpful and relevant.

THE ANSWER KEY WILL BE POSTED Nov. 21, 2025.

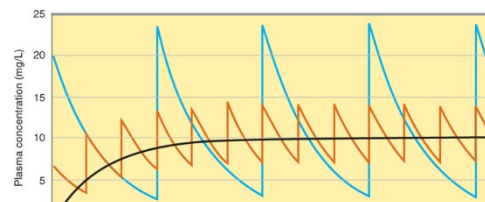
For multiple-choice questions, explain why the incorrect options are not correct. In all cases, explaining the answers (out loud) will help solidify the learning.

First, the basics

1. Rational dosing applies the dose-effect relationship. What are two goals of this fundamental approach to safe drug therapy for the individual patient?

Hint: Perhaps the figure will give you a clue.

Short answer.



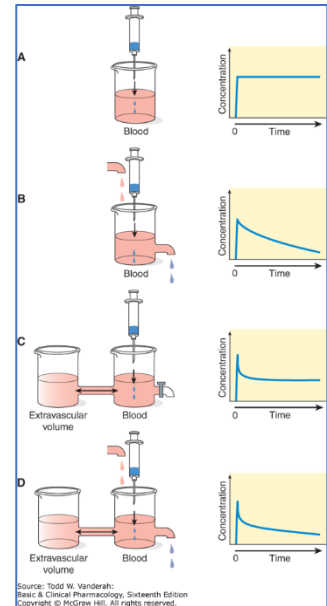
Basic and Clinical Pharmacology Figure 3-6

Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition
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2. What are the 4 most important parameters governing the concentration, movement, and time course of drug effect in the body, collectively known as drug disposition?

Hint: Perhaps the simple model at the right will help you. (Notes p1)

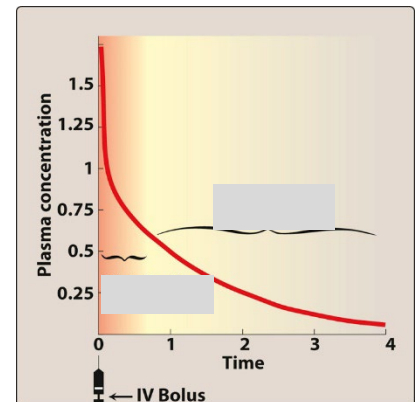
Short answer



Basic & Clinical Pharmacology
Figure 3-2

3. A patient is given an intravenous bolus of a drug that achieves a peak serum concentration of 1.75 mcg/mL. The plasma concentration vs. time curve shows a steep decline initially, then a more gradual decline.

- What is the explanation for the biphasic concentration-time curve?
- What are the 2 phases of the curve called?



4. A 37-year-old woman with major depressive disorder is started on a standard low dose of an antidepressant. The doctor explains that the antidepressant effect usually take several weeks. Therapy will be started with a low dose that will be gradually increased to optimal therapeutic effect based on response and on tolerability. The drug has a $t_{1/2}$ of 7 days.

- What is the approximate time to steady state concentration?
 - 7 days
 - 14 days
 - 28 days
 - 70 days
- How often should a dosage increase be considered, if needed?
 - 7 days
 - 14 days
 - 28 days
 - 70 days

Please explain your reasoning.

5. A 76-year-old man is hospitalized for the management of acute ischemic stroke and blood pressure of 195/112 mmHg after reperfusion therapy. The patient is given an IV loading dose of labetalol (a beta blocker) followed by continuous intravenous infusion of the maintenance dose of the drug to reduce his blood pressure to a target of <180/105 mmHg in the first 24 hours. Labetalol has a half-life of about 5.5 hours, according to information in the drug monograph. What is the purpose of this dosing strategy?

6. The patient in the above vignette is given labetalol continuous infusion at 5 mg/minute. The dosing rate calculation included the value 105 L/h for a 70 kg adult obtained from a table. This parameter is defined as the volume of blood plasma (105 liters) removed of drug by elimination over a specific time period (1 hour) in a 70 kg adult.

Which of the following is defined by this parameter?

- A. Area under the concentration-time curve
- B. Clearance
- C. Steady state
- D. Volume of distribution

Define each term.

7. A critically ill patient is administered the antibiotic gentamicin for a severe gram-negative infection. Dosing calculation included the V_d value 0.25 L/kg.

What does this value indicate about the drug's distribution?

Short answer

Hint: Notes p10

8. A normally healthy 56-year-old man is hospitalized for treatment of a resistant staphylococcal (MRSA) infection with I.V. vancomycin (antibiotic). Plasma sampling is drawn for AUC-based dosing.

Which of the following relationships BEST reflects this parameter selected for optimizing vancomycin dosing in this patient?

- A. The dose in relation to the plasma concentration
- B. The plasma concentration in relation to the blood flow
- C. The rate of drug elimination in relation to plasma concentration
- D. The time required to reduce the plasma concentration by one-half
- E. The variation in plasma concentration over time

9. A 65-year-old woman is brought to the emergency department because she experienced a seizure at home. History of seizure disorder is negative. Medical history is significant for type 2 diabetes mellitus and chronic kidney disease. She receives an intravenous infusion of the antiseizure drug levetiracetam.

Levetiracetam properties:

- V_d : ~ 0.5 L/kg
- plasma protein binding $< 10\%$ (minimal)
- Not appreciably metabolized
- Excreted in urine as unchanged drug

Which of the following options should be anticipated with the use of this drug in this patient?

	Half-life	Dosage
A.	increased	increased
B.	decreased	decreased
C.	increased	no change
D.	increased	decreased
E.	no change	no change

Definition – chronic kidney disease: Reduced ability of the kidney to filter the blood efficiently
glomerular filtration rate $\rightarrow$ reduced glomerular filtration rate

10. A 64-year-old female patient (60 kg) is treated with experimental Drug A for type 2 diabetes. Drug A is available as tablets with an oral bioavailability of 90%. If the V_d at steady state is 2 L/kg and the desired steady-state plasma concentration is 3.0 mg/L, which of the following is the most appropriate oral loading dose of Drug A?

- A. 6 mg
- B. 6.66 mg
- C. 108 mg
- D. 360 mg
- E. 400 mg

11. A 45-year-old patient with small cell lung cancer has elected to participate in the trial of a new chemotherapy drug. It is given by continuous IV infusion at a rate of 10 mg/h. Plasma concentrations (C_p) are measured. Results are on the table at the right.

Time after start of the infusion	C_p in mg/L
1	0.7
4	3.0
8	3.6
16	3.84
24	4.0
30	4.0

What conclusion can be drawn from these findings?

- A. Volume of distribution is 30 L.
- B. Clearance is 2.0 L/h.
- C. Elimination follows zero-order kinetics.
- D. Half-life is 2 hours.
- E. Doubling the rate of infusion would result in plasma concentration of 16 mg/L at 40 hours.

12. Recall questions:

i) Which one of the following equations may be useful for modeling the effect of a diseased metabolizing organ on a drug's clearance?

A. $E = CL / Q$

B. $CL = (0.693 \times V_d) / t_{1/2}$

C. $F = AUC \text{ (oral)} / AUC \text{ (I.V.)}$

D. $K_a = 1 / K_d$

ii) What do the other options represent.

13. A frail 87-year-old woman with a long history of hypertension and atrial fibrillation develops dyspnea and fatigue of 1-week duration. The patient lives at home alone. A home health aid comes in four days a week for four hours a day. Her son visits every day and her daughter visits once a week.

VS: Pulse 112 bpm, BP 188/72 mmHg, RR 23 breaths per minute, BMI 21 kg/mm².

PE: Jugular venous distention, hepatomegaly, pulmonary crackles, S1, S2, S3 heart sounds, and pedal edema are noted

Labs: Samples for complete blood count, electrolytes, cardiac troponin, BUN, creatinine, and liver function tests are obtained.

Imaging: Chest radiograph shows cardiomegaly and evidence of pulmonary edema. EKG and Doppler echocardiogram are consistent with congestive heart failure with reduced ejection fraction.

The patient is admitted for management of acute decompensated heart failure.

In view of the patient's advanced age, frailty, and clinical condition, what are the potential alterations in the pharmacokinetic properties?

What are the clinical implications for pharmacotherapy in this patient?

Hint: Notes p23-24 (Please try to avoid the temptation to copy and paste the answers.)

Pharmacokinetics changes in the older patient with heart failure	
oral absorption	
volume of distribution	
plasma protein binding	
metabolism (hepatic clearance)	

bioavailability	
renal excretion	
half-life	

14. JP is a 7-year-old male diagnosed with focal seizures with secondary generalization. He is initiated on a standard starting dose of phenytoin. His mother is directed to have her son thoroughly chew 1 tablet 3 times daily (Sig: Chew 1 tab TID). He can take it with food if it bothers his stomach. The doctor also talks to JP to make sure he understands why he is getting the medicine and how to take it and that the tablets taste minty.

JP returns to the neurologist in 7 days for evaluation. His phenytoin trough plasma concentration is 6 mcg/mL (therapeutic range 10-20 mcg/mL). He has not had a seizure event during this time and has been taking the medication exactly as directed. The physician wants to double the serum concentration.

If the dose of phenytoin is doubled, what serum concentration could be anticipated?

- A. The C_p will double because of linear elimination kinetics.
- B. The C_p may significantly increase because of nonlinear elimination kinetics.

15. A hospitalized patient is treated for supraventricular arrhythmias with verapamil continuous I.V. infusion at 10 mg/h. When stable, she is switched to an oral extended-release verapamil capsule at 1 cap every 24 hours. The patient's weight is 70 kg.

Which one of the following calculations would be useful in predicting the dose of a once daily extended-release capsule to maintain the drug concentration within the therapeutic window?

A. $\frac{(CL \times C_p) \times 24 \text{ h}^*}{F}$	B. $\frac{(V_d \times C_p) \times 24 \text{ h}}{F}$	C. $\frac{(CL \times AUC) \times 24 \text{ h}}{F}$	D. $\frac{24 \text{ h AUC-IV}}{24 \text{ h AUC-oral}}$
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16. A 44-year-old male presents to the ED with a blood pressure of 185/112 mmHg. He is treated with an I.V. antihypertensive agent.

Which of the following is the preferred method of monitoring the effect of this drug?

- A. Blood pressure measurement
- B. Cardiac telemetry
- C. EKG
- D. Therapeutic blood levels

17. A generic drug manufacturer is developing a version of a popular brand-name antihypertensive tablet, whose patent has expired. The company submits to the FDA data demonstrating that:

- the drug contains the identical active ingredient in identical amounts as the reference drug,
- the rate and extent of absorption do not show a significant difference from the rate and extent of absorption of the reference drug, and
- the generic version can be expected to have the same clinical effect and safety profile as the reference drug when administered to patients under conditions specified in the labeling.

Which one of the following is the generic drug maker demonstrating to the FDA?

- A. The generic version is a bioequivalent.
- B. The generic version is a pharmaceutical equivalent.
- C. The generic version is a therapeutic equivalent.
- D. The generic version is a biosimilar.
- E. The generic version is an interchangeable biological.

18. A pharmaceutical company is developing a mint-flavored orally disintegrating tablet of their popular antihistamine oral tablet. It rapidly dissolves when placed on the tongue and can be taken without water.

Which of the following parameters is most useful in comparing both the rates and extents of bioavailability of the company's oral tablet and the orally disintegrating tablet of their drug?

- A. AUC
- B. Bioavailability
- C. CL
- D. $t_{1/2}$
- E. V_d